Price Elasticity of Demand

how fast buyers react!

Business Economics

In simple words *Elasticity tells us by what percent demand changes when price changes by 1 percent.*

1 Price elasticity of demand

Responsiveness of quantity demanded of a good to a change in its own price, other factors remaining constant.

Ceteris paribus always!

2 Elastic vs inelastic

Elastic: |Ed| > 1, quantity changes proportionately more than price. Inelastic: |Ed| < 1, quantity changes proportionately less.

Unitary: |Ed| = 1

1. Ed is a pure number: it has no units, so it does not depend on ₹ or kg.
2. For a normal downward-sloping demand curve, Ed is negative; we usually read its absolute value.
3. Necessities tend to be inelastic; luxuries and goods with many substitutes tend to be elastic.
4. Perfectly elastic = infinity (horizontal curve); perfectly inelastic = 0 (vertical curve).



Swipe for the formula →

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THE ED FORMULA

Price Elasticity of Demand (PED)

percent over percent!

Percentage method

$$E_D = \frac{\%\ \text{change in quantity demanded}}{\%\ \text{change in price}}$$

Ed = price elasticity of demand; quantity demanded is of the same good; sign is negative on a normal demand curve.

Change-in-values form

$$E_D = \frac{\Delta Q \times P}{\Delta P \times Q}$$

Q = original quantity, P = original price, ΔQ = change in quantity, ΔP = change in price.

Total outlay link

E_D = Price up: outlay falls if $|Ed|>1$, rises if <1, same if $=1$

Outlay = P × Q, which equals the seller's total revenue.

ELASTIC: PRICE AND OUTLAY MOVE OPPOSITE! If $|Ed| > 1$, a price rise cuts total revenue; if $|Ed| < 1$, the same price rise increases total revenue.

Swipe for a memory trick ⟶

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STUDY TRICK

Price Elasticity of Demand (PED)

remember the three cases

ELASTIC = ESCAPES QUICKLY

Elastic buyers escape when price rises, so revenue falls. Inelastic buyers stay stuck, so revenue rises. Unitary buyers: revenue stays same.

HOW TO ATTEMPT IT

1. Find ΔQ and ΔP from the data →
2. Use original Q and original P →
3. Apply ΔQ/Q ÷ ΔP/P →
4. Ignore the sign; compare with 1 →
5. Name it: elastic, unitary or inelastic

EXAM TIP

In MCQs, first compare |Ed| with 1, then match the revenue effect. Check whether the question gives a price rise or fall.

DOUBT BUSTER

Why is a flat demand curve not simply 'more elastic' at every point? Only a horizontal line is perfectly elastic. For a straight-line demand curve, Ed changes from point to point, high at top and low at bottom.

Swipe for common mistakes →

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COMMON MISTAKES

Price Elasticity of Demand (PED)

avoid these slips!

✗ Elasticity is the same as the slope of the demand curve

✓ Slope is $\Delta P/\Delta Q$ in absolute units; Ed uses percentage changes, so it is unit-free and changes along a straight line.

✗ Perfectly inelastic demand means Ed = infinity

✓ Perfectly inelastic means Ed = 0 (vertical curve); perfectly elastic means Ed = infinity (horizontal curve).

✗ If $|Ed| < 1$, a price rise reduces total revenue

✓ If $|Ed| < 1$ (inelastic), a price rise increases total revenue because quantity falls proportionately less.

✗ Using the new price and quantity in the denominator

✓ In the basic formula, use the original price P and original quantity Q unless the question asks for the arc method.

TOMORROW'S CA FOUNDATION MCQ TESTS THIS! Revise it tonight, then try the MCQ.

Save this for revision!

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